Supplement: empirical probes and cross-disciplinary reframings of the Goldbach–Chen frontier

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This supplement to *The Goldbach–Chen Frontier: A Singular-Series Cancellation behind a Finite-Range Half-Power* has three parts. Section 1 distils, from the paper's cross-disciplinary reframings, the one part with genuine mathematical content—a structural *negative*—and disposes of the rest in a sentence. Section 2 collects the empirical probes moved out of the body for length. Section 3 gives two *deferred analytic models*—conditional propositions, with proofs—that explain two of those probes. "The main paper" refers to the article above; because it is a separate document, cross-references to it are made by name (e.g. "Theorem 2 of the main paper"), and every number here is reproducible from the `run_*.py` drivers in the repository.

1 An honest structural negative

The frontier admits many external recastings—thermodynamic (an arithmetic ground state), network (adversarial robustness), control (value of information), information-theoretic, and geometric—but each is the same fact, that $\mathfrak{S}(N)$ sets the local densities, seen in a different mirror, with no result of its own; we do not develop them (all are reproducible from the `run_*.py` drivers). One recasting does return a genuine *structural negative*, worth recording because it bounds what the additive structure does *not* buy, and we give it in full so the claim can be checked.

Persistent topology of additive decompositions. Filter the binary decompositions of N by *arithmetic complexity*: form the simplicial complex

$$K_N^{(k)} = \big\{ \{a,b\} \ :\ a+b=N,\ \ \Omega(a)+\Omega(b)\le k \big\}, \qquad k=2,3,\dots,$$

whose vertices are the summands a and whose edges are the decompositions admitted once their total complexity $\Omega(a)+\Omega(b)$ is at most k; growing k gives the sublevel filtration. Because each integer a with $0<a<N$ lies in *exactly one* pair $\{a,N-a\}$, every $K_N^{(k)}$ is a *disjoint matching* (every vertex has degree ≤ 1): it contains no path of length two, hence no 1-cycle and no higher simplex, so $H_i\big(K_N^{(k)}\big)=0$ for all $i\ge 1$ and all k. Its 0-dimensional persistence is therefore exactly the energy spectrum—each pair $\{a,N-a\}$ is a connected component born at its energy $\Omega(a)+\Omega(N-a)$ and never merging with another. The complexity filtration thus adds *nothing* beyond the spectrum; we record it as a clean negative. Genuine higher-dimensional structure appears only at the *ternary* level—the graph on primes $p,q<N$ with an edge whenever $N-p-q$ is prime is connected (a graph form of Helfgott's ternary theorem) but strongly *anti*-clustered, with far fewer triangles than a random graph of equal density, so even there the representations do not aggregate into higher simplices.

2 Further empirical probes (moved from the main text)

For length and focus, the following experiments were moved out of the main text. Each is reproducible from its `run_*.py` driver in the repository and, like every reframing here, returns to the same singular series $\mathfrak{S}(N)$.

2.1 The Chen-rescue threshold

Restricting representations to a balanced band $p \in [N/2-k,\, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 1). For $k \gtrsim 0.005\,N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at $k \sim 20$–40; for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), where the expected number of balanced prime pairs $\sim 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$ drops to $O(1)$. This is the regime captured by the residual exponent $a_{\mathrm{Res}}(N)$ of the main text: Chen's flexibility becomes *necessary* precisely when near-perfect balance is imposed, so the rescue threshold and a_{Res} track the same singular series.

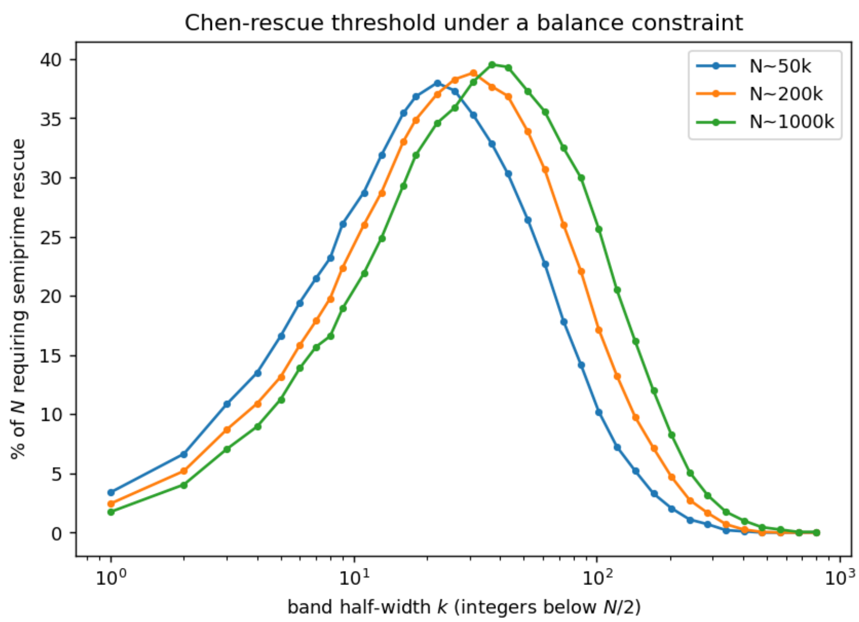

Figure 1: Fraction of N that require a semiprime rescue versus band half-width k, for three scales of N. The curve is unimodal with peak scaling like $\log^2 N$.

2.2 Weak continuity of the location measures

The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X+H]$ produces a stable density (Fig. 2). At $X = 10^6$: *(i)* splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (a concrete form of weak convergence); *(ii)* the prime-q density is within total variation 0.028 of uniform on $(0,1)$, with a mild endpoint upturn (the $1/(\log(xN)\log((1-x)N))$ effect); *(iii)* prime and semiprime locations agree to total variation 0.019, the semiprime location slightly skewed toward larger q (mean $q/N = 0.513$ versus 0.500 for primes). This probe earns its place because it is explained by the deferred analytic model of §3.1 below: under the weighted Hardy–Littlewood asymptotics, after normalization both the inherited singular series *and* the Mertens channel sum cancel, leaving the uniform law for q/N in either channel—the same cancellation that drives the whole paper, now visible in the location law.

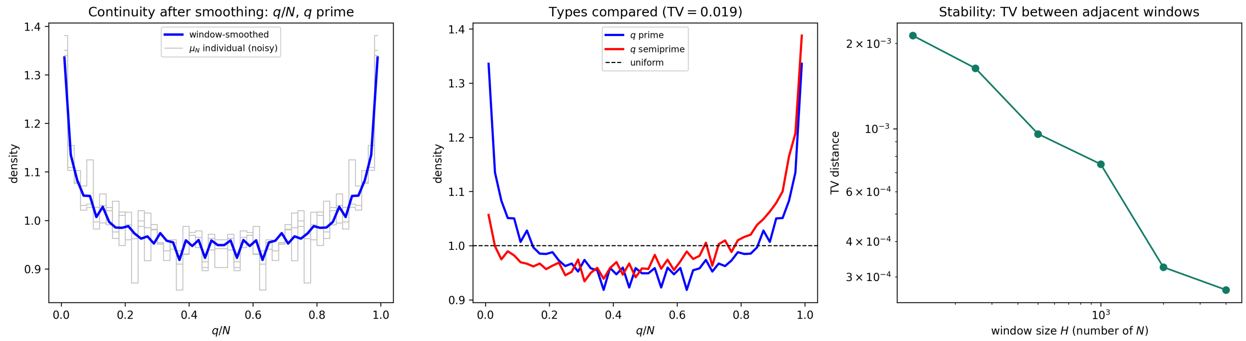

Figure 2: Weak continuity of q/N. Left: noisy pointwise μ_N versus the window-smoothed density. Center: prime versus semiprime location densities, both near uniform. Right: same-scale sampling distance versus window size H (log–log).

2.3 Topology of valleys

Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\le 2}/(\mathfrak{S}+\mathfrak{S}^{1/2}W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at $+0.64$, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 3); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime. The correlation $+0.64$ is a within-window descriptive statistic over a strongly arithmetically dependent sequence (consecutive even N), not an i.i.d. estimate.

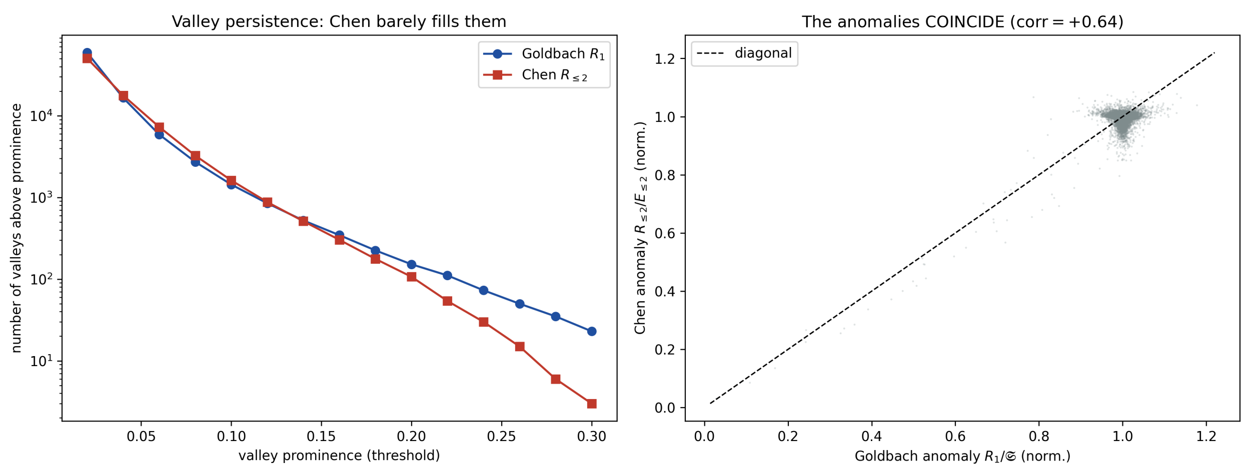

Figure 3: Valley persistence. Left: Chen has nearly as many valleys as Goldbach. Right: the normalised anomalies coincide (+0.64).

2.4 Statistical dynamics of the frontier

The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding $N \bmod 30$ contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \to \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (lag-1 autocorrelation -0.21 after detrending): the frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 4). These percentages are descriptive R^2 values over an arithmetically dependent window, not i.i.d. fits.

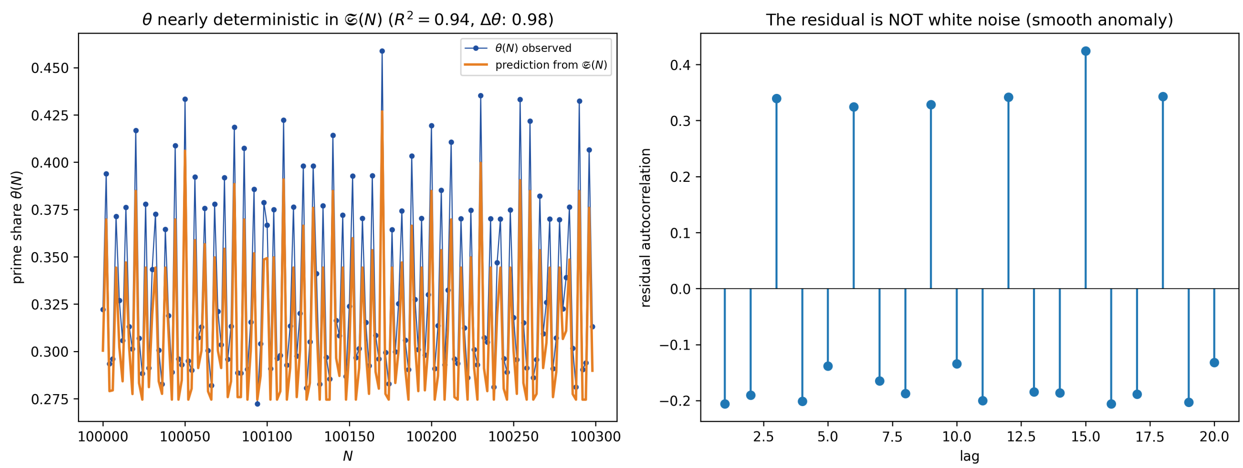

Figure 4: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.

3 Deferred analytic models

These two conditional model statements were moved out of the main paper's appendix—which it keeps to three parts (the identity, the unconditional almost-all asymptotic, and the conditional program $\beta_2 \to 1$)—into this supplement. They explain, respectively, the weak-continuity probe of §2.2 and the carry signature of the main paper. Like the model statements that remain in the main paper's Appendix C, each holds in a stated model, not unconditionally.

3.1 Weak uniformity of the location measures

The continuity experiments of §2.2 suggest that the representation clouds become continuous after normalization: the empirical laws of q/N for both prime and semiprime q approach the uniform law on $(0,1)$, up to small endpoint effects. This is what the weighted Hardy–Littlewood heuristic predicts. For a bounded C^1 test function $\varphi : [0,1] \to \mathbb{R}$ write, when $R_t(N) \neq 0$,

$$\mu_N^{(1)}(\varphi) := \frac{1}{R_1(N)} \sum_{\substack{p<N \\ p,\, N-p \in \mathbb{P}}} \varphi\Big(\tfrac{N-p}{N}\Big), \qquad \mu_N^{(2)}(\varphi) := \frac{1}{R_2(N)} \sum_{\substack{p<N \\ p \in \mathbb{P},\, N-p \in \mathcal{S}_2}} \varphi\Big(\tfrac{N-p}{N}\Big),$$

the averages of φ against the type-t location measures.

Hypothesis 1 (Weighted prime and semiprime channel asymptotics)**.** *For every bounded C^1 test function φ and almost all even N,*

$$\sum_{\substack{p<N \\ p,\, N-p\in\mathbb{P}}} \varphi\Big(\tfrac{N-p}{N}\Big) = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{N}{\log^2 N} \int_0^1 \varphi(x)\, dx,$$

and, for each unblocked semiprime channel $q = rs$, uniformly in the range of r used in the paper,

$$\sum_{\substack{s\in\mathbb{P} \\ N-rs\in\mathbb{P}}} \varphi\Big(\tfrac{rs}{N}\Big) = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{r-1}{r-2}\, \frac{N}{r\log^2 N} \int_0^1 \varphi(x)\, dx,$$

the channel omitted when $r \mid N$.

Proposition 1 (Weak uniformity of the location measures; conditional)**.** *Under Hypothesis 1, for almost all even N,*

$$\mu_N^{(1)}(\varphi) = \int_0^1 \varphi(x)\, dx + o(1), \qquad \mu_N^{(2)}(\varphi) = \int_0^1 \varphi(x)\, dx + o(1).$$

Equivalently, the empirical measures of q/N in the prime and semiprime channels converge weakly to the uniform law on $(0,1)$, in the same smoothed/almost-all sense as the weighted asymptotics.

Proof. For the prime channel, the first weighted asymptotic with $\varphi \equiv 1$ gives $R_1(N) = (1 + o(1))\, 2C_2\mathfrak{S}(N)N/\log^2 N$; dividing the two estimates yields $\mu_N^{(1)}(\varphi) = \int_0^1 \varphi + o(1)$. For the semiprime channel, writing $q = rs$ and summing the per-channel asymptotic over the unblocked $r \le \sqrt{N}$,

$$\sum_{\substack{p<N \\ p\in\mathbb{P},\, N-p\in\mathcal{S}_2}} \varphi\Big(\tfrac{N-p}{N}\Big) = (1+o(1))\, 2C_2\mathfrak{S}(N)\, \frac{N}{\log^2 N}\Big(\sum_{\substack{3\le r\le\sqrt{N} \\ r\nmid N}} \frac{r-1}{r-2}\frac{1}{r}\Big) \int_0^1 \varphi(x)\, dx,$$

the parenthesis being the Mertens channel sum $W_f(N)$. Taking $\varphi \equiv 1$ gives the matching asymptotic for $R_2(N)$; dividing cancels both the inherited singular series $\mathfrak{S}(N)$ and the channel sum, leaving $\mu_N^{(2)}(\varphi) = \int_0^1 \varphi + o(1)$. This is weak convergence against every bounded C^1 test function. □

This explains why the discontinuous pointwise representation sets nonetheless produce smooth limiting clouds: the singular series governs the local *density* of representations, but after normalizing each measure it cancels—together with the Mertens channel sum in the semiprime case—leaving the uniform spatial law. It is the conditional counterpart of the weak-continuity diagnostics of §2.2, and like the carry model of §3.2 below it is conditional on the stated weighted asymptotics, not an unconditional theorem about primes.

3.2 A last-digit model for the carry excess

The carry geometry observed in the main paper can be partly explained by a local digit calculation: prime-like summands are not uniform among terminal digits but restricted to reduced residue classes, and this restriction alone creates a positive carry bias in the least significant digit. Fix a base $b \ge 2$ and let $U_b := \{1 \le u \le b-1 : (u,b) = 1\}$ be the reduced residue digits modulo b. For digits $x, y \in \{0, \dots, b-1\}$ let $\kappa_b(x,y) := \mathbf{1}_{\{x+y\ge b\}}$ be the carry out of the least significant digit.

Proposition 2 (Local carry excess; model)**.** *Let X, Y be independent uniform digits in $\{0, \ldots, b-1\}$ and U, V independent uniform digits in U_b. Then*

$$\mathbb{E}\,\kappa_b(U,V) = \tfrac{1}{2} + \tfrac{1}{2}\Pr(U+V=b), \qquad \mathbb{E}\,\kappa_b(X,Y) = \frac{b-1}{2b},$$

so $\mathbb{E}\,\kappa_b(U,V) - \mathbb{E}\,\kappa_b(X,Y) = \frac{1}{2b} + \frac{1}{2}\Pr(U+V=b) > 0$: reduced-residue summands have a strictly larger least-digit carry probability than unrestricted ones.

Proof. For unrestricted digits, $\#\{(x,y) : 0 \le x, y \le b-1,\ x+y \ge b\} = \sum_{x=1}^{b-1} x = \frac{b(b-1)}{2}$, so $\mathbb{E}\,\kappa_b(X,Y) = (b-1)/(2b)$. For reduced residues, the involution $(u,v) \mapsto (b-u, b-v)$ maps U_b^2 to itself (since $(u,b)=1 \iff (b-u,b)=1$), sends $\{u+v<b\}$ bijectively to $\{u+v>b\}$, and fixes $\{u+v=b\}$; hence those two strict sets are equinumerous and $\Pr(U+V \ge b) = \frac{1}{2} + \frac{1}{2}\Pr(U+V=b)$. $\square$

The calculation is deliberately local: it does not prove the full carry statistics of actual Goldbach pairs, nor assume independence of higher digits. It shows that the most basic prime residue restriction already pushes the carry count up. In base 2 both odd prime summands end in 1, so the least digit always carries; in larger bases the effect survives after averaging over reduced classes. Since the total carry count obeys the Kummer–Legendre identity $\#\mathrm{carries}_b(a,q) = (S_b(a)+S_b(q)-S_b(N))/(b-1)$ (here S_b is the base-b digit sum, as in the main paper's carry observation), the proposition is a positive *local* contribution to it, and the empirical carry excess reported in the main paper reads as its multi-digit continuation.

Remark 1 (Fixed-N caveat)**.** *For a fixed terminal digit $N \bmod b$ the carry bias can depend on the residue class and need not be positive in every class; the proposition is an averaged local-prime model, explaining why a positive excess is natural* on average *while the full statistic remains a global multi-digit measurement. Like the weak-uniformity model of §3.1 above, it is model-based, not a theorem about the empirical carry law.*